SEQUENCES LADDER 3 OF 9

Cubics and higher differences

Differences of a degree k rule settle at the constant $k!$ times the leading coefficient.

1 / THE RULE AND THE PICTURE

A rule of degree k has constant k-th differences equal to $k!\,a$

u_n	4	15	48	115	228
Δ	11	33	67	113	
Δ^2	22	34	46		
Δ^3	12	12			

For the terms $4, 15, 48, 115, 228$ the third differences are the constant 12, and

$$\Delta^k u_n = k!\,a_k \quad \text{for degree } k$$

so $a_3 = 12/3! = 2$ and the rule starts $2n^3$. The same peeling as for quadratics finishes the job.

2 / PEELING OFF ONE DEGREE AT A TIME

Subtract the leading term and look at what is left

Subtracting $2n^3 = 2, 16, 54, 128, 250$ from $4, 15, 48, 115, 228$ leaves $2, -1, -6, -13, -22$, whose second differences are the constant -2, so $a = -1$ and the next term is $-n^2$.
Adding $n^2 = 1, 4, 9, 16, 25$ gives $3, 3, 3, 3, 3$, so the rule is $u_n = 2n^3 - n^2 + 3$. Each pass removes one power, and the process stops when a constant is left.

3 / WHY $k!$ APPEARS

Each difference lowers the degree and multiplies by the old degree

degree	rule	constant difference
degree 1	$u_n = an + b$	$\Delta u = a = 1!\,a$
degree 2	$u_n = an^2 + \dots$	$\Delta^2 u = 2a = 2!\,a$
degree 3	$u_n = an^3 + \dots$	$\Delta^3 u = 6a = 3!\,a$
degree k	$u_n = an^k + \dots$	$\Delta^k u = k!\,a$

The step from n^k to $(n+1)^k$ is $kn^{k-1} + \dots$, a polynomial of degree $k-1$ with leading coefficient k, and repeating k times multiplies the coefficients $k(k-1)\cdots 1 = k!$.

4 / A SUM THAT TURNS OUT CUBIC

Adding up squares

The running totals of the squares $1, 5, 14, 30, 55, 91$ have first differences $4, 9, 16, 25, 36$, second differences $5, 7, 9, 11$ and third differences $2, 2, 2$, so the totals follow a cubic with leading coefficient $2/3! = \frac{1}{3}$, which matches $\frac{1}{3}n^3$ in the formula $\frac{n(n+1)(2n+1)}{6}$ for the sum of the first n squares.

5 / HOW MANY TERMS ARE ENOUGH

One more than the number of unknowns

A cubic has four coefficients, and checking that the third differences are constant needs two of them, so 5 terms are the least that confirm a cubic.
If the differences never settle, as for $1, 2, 4, 8, 16$ whose differences are $1, 2, 4, 8$ again, the rule is not a polynomial, and the last sheet shows it is exponential.

6 / TRY THESE

Find the degree and the leading coefficient

(a) For $1, 8, 27, 64, 125, 216$ the third differences are 6, so $a = 6/6 = 1$ and the rule is n^3. **(b)** For $1, 16, 81, 256, 625$ the fourth powers have fourth differences 24, so $a = 24/4! = 1$. **(c)**
For $3, 10, 29, 66, 127$ the third difference is 6 and subtracting n^3 leaves $2, 2, 2, 2, 2$, so $n^3 + 2$.

ASIDE / THE FACTORIAL

The factor $k!$ is the same one that appears when a power is differentiated k times, $\frac{d^k}{dx^k} x^k = k!$.

LINKS TO OTHER SHEETS

The next sheet turns the difference table into a formula, and later sheets compare differences with derivatives.

Convention. Δ^k is the k-th difference; a_k is the coefficient of n^k; $k! = k(k-1)\cdots 1$.